

NEW PRODUCTS

TECHNOLOGY

COMPONENTS

Bipolar switching devices

Designers building modern electric vehicles face a tough trade-off between ever-higher power and tighter



space and heat budgets. To boost efficiency and reliability in high-power automotive electronics, Diodes Incorporated

has introduced its DXTN/DXTP 78Q and 80Q bipolar switching devices, featuring ultra-low saturation voltages and high current capability. These parts are aimed at gate driving for MOSFETs/IGBTs, DC-DC conversion, power and load switching, motor drives, solenoids and relay/actuator control in EV powertrains and ancillary systems.

Diodes Incorporated

<https://www.diodes.com>

CAN chips

Holtek has introduced a family of 5V compatible CAN/LIN-enabled ARM Cortex-M0+ microcontrollers designed



to simplify networking and control in 5V embedded systems without the

need for external level shifters. The HT32F53231/41/42/52 microcontrollers integrate a Bosch-licensed CAN 2.0A/B controller with LIN support and operate from 2.5-5.5V, covering applications in industrial automation, building systems, electric two-wheelers, and on-board diagnostics.

Holtek

<https://www.holtek.com/page/index>

Violet laser diode

Nuvoton's latest 402nm high-power violet laser diode is emerging as a solid-state alternative to traditional

mercury lamps by delivering around 1.7W optical output in a compact



TO-56 CAN package, easing heat and size constraints that have

long limited violet/blue-light sources. With enhanced reliability and smaller form factor, these lasers suit laser direct imaging, resin curing, welding, 3D printing, and biomedical optics, replacing bulky mercury bulbs and enabling tighter, more efficient optical designs.

Nuvoton Technology

<https://www.nuvoton.com>

Mobile processors

Intel's new Core Ultra Series 3 processors are built on its advanced 18A



process to meet rising demand for on-device AI and high-performance

computing across edge systems. Targeting laptops, robotics, smart cities, automation, and healthcare, the chips combine multicore CPUs, Arc graphics, and AI acceleration (up to ~50 TOPS) to boost inference, video analytics, and vision-language workloads locally without relying on the cloud. They support extended battery life, certified embedded use, and richer AI features at the edge.

Intel

<https://www.intel.com>

Hall effect current sensors

Littelfuse has launched a new family



of automotive-qualified Hall-effect current sensors tailored to elec-

tric and hybrid vehicles, addressing precise, isolated measurement needs in

battery management, motor control, and safety circuits. The portfolio spans $\pm 1500A$ ranges, offers analogue or CAN/LIN outputs, and includes ASIL-capable models with diagnostics plus a fast pyro-fuse trigger option. These sensors support compact busbar integration and functional safety for EV powertrain and protection systems.

Littelfuse, Inc.

<https://www.littelfuse.com>

Smart fan driver

Melexis' new MLX90411-D smart fan driver helps systems cut cooling noise



and improve fan lifetime by stabilising low-speed operation

and offering flexible speed curves for precise airflow control. It is aimed at consumer electronics such as gaming PCs and consoles, home appliances such as refrigerators, and industrial power systems, including UPS and energy storage, where quieter, efficient cooling, and simpler integration matter most.

Melexis

<https://www.melexis.com/en>

DC motor driver ICs

Designers targeting compact electrified systems can now use ROHM's



new BD60210FV and BD64950EFJ brushed DC motor

driver ICs to streamline motion control across everyday products. The dual- and single-channel drivers support H-bridge control with PWM/constant-current options, cutting external parts and idle power. They suit home appliances (refrigerators, air-conditioners, printers, robot vacuums) and light industrial equipment (automatic doors, conveyors, power tools).

Rohm Semiconductor

<https://www.rohm.com>